

Lidocaine

(Xylocaine)

I. Classification – Antiarrhythmic, anesthetic**II. Action(s)**

- Class 1B antiarrhythmic
- Depresses Phase 0 of depolarizing action potential
- Stabilizes electrically charged cell membrane (cardiac, nerve) slowing activity

III. Indication(s)

- Ventricular arrhythmias (Consult Medical Direction for administration)
- Local anesthetic
- Non-EMS use: anesthesia / RSI adjunct (CNS depressant)

IV. Contraindications

- Known Hypersensitivity
- Wolff-Parkinson-White syndrome (WPW)
- Heart Block (without implanted pacemaker)

V. Precautions

- May significantly worsen heart blocks and slow AV nodal conduction
- CNS depressant

VI. Drug interactions

- Amiodarone and Beta-blockers – Significant QT prolongation, hypotension
- Use with caution with other CNS depressants

VII. Adult Dosing and Administration

- VF / Pulseless VT: 1 mg / kg IO / IV initial dose, repeat doses 0.5 mg / kg every 5 to 10 minutes (max total dose 3 mg / kg)
- Stable VT: 1 mg / kg (max dose 100mg) IO / IV, repeat doses 0.5 mg / kg every 5 to 10 minutes (max total dose 3 mg / kg) or infusion after initial bolus of 1-4mg/min titrated to rhythm response (max cumulative dose within 1hr = 300mg)
- EZ-IO Infusion Pain: 0.5 mg / kg IO (max 40 mg or 2 mL Lidocaine 2%) over 2 minutes, allow to dwell for 60 seconds, flush with 2-5 mL NS

VIII. Pediatric Dosing and Administration

- Stable VT: 1 mg / kg (max dose 100mg) IO / IV, repeat doses 0.5 mg / kg every 5 to 10 minutes (max total dose 3 mg / kg) or infusion after initial bolus of 1-4mg/min titrated to rhythm response (max cumulative dose within 1hr = 300mg)
- EZ-IO Infusion Pain: 0.5 mg / kg IO (max 40 mg) over 2 minutes, allow to dwell for 60 seconds, flush with 2-5 mL NS

IX. Onset

- 1-5 minutes

X. Duration

- 1.5 – 2 hours

XI. Possible Adverse Effects

- Heart block / Hypotension
- CNS Depression
- Ventricular Arrhythmias
- Numbness at application site
- Respiratory depression / status asthmaticus